

JKBOSE 2026

CLASS 11TH

**TRIGONOMETRIC
FUNCTIONS**

MATHS

LECTURE-04

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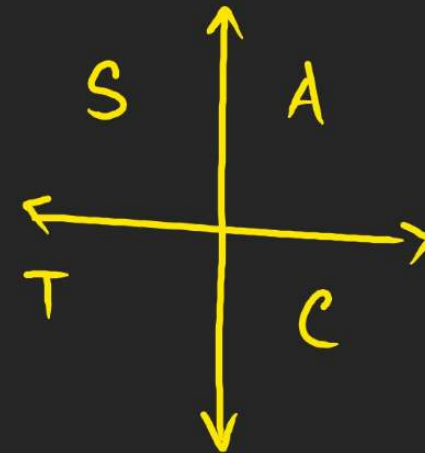
TOPICS TO BE COVERED

1

Exercise Discussion 3.2

2

Related Problems



QUESTION

If $\sec\theta + \tan\theta = \frac{1}{4}$ then θ lies in the

~~1~~ First Quadrant

~~2~~ Second Quadrant

~~3~~ Third Quadrant

☒ 4 Fourth Quadrant

(Sol) Given $\sec\theta + \tan\theta = \frac{1}{4}$ — (1)

We know that,

$$\sec^2\theta = 1 + \tan^2\theta$$

$$\Rightarrow \sec^2\theta - \tan^2\theta = 1$$

$$\Rightarrow (\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1$$

$$\Rightarrow \frac{1}{4} (\sec\theta - \tan\theta) = 1$$

$$\Rightarrow \sec\theta - \tan\theta = 4 \text{ — (2)}$$

Adding (1) and (2), we get;

$$2\sec\theta = \frac{1}{4} + 4 = \frac{17}{4}$$

$$\Rightarrow \sec\theta = \frac{17}{8} > 0$$

Subtracting (1) and (2), we get;

$$2\tan\theta = \frac{1}{4} - 4 = -\frac{15}{4}$$

$$\Rightarrow \tan\theta = -\frac{15}{8} < 0$$

$\Rightarrow \theta$ lies in the IV Quadrant.

QUESTION

Which of the following is not correct?

~~1~~ $\sin \theta = -\frac{1}{5}$

~~2~~ $\cos \theta = 1$

☒ 3 $\sec \theta = \frac{1}{2}$

~~4~~ $\tan \theta = 20$

Range of T-functions:

Range of Sine = $[-1, 1]$

Range of Cosine = $[-1, 1]$

Range of Tangent = $\mathbb{R}$

Range of Cotangent = $\mathbb{R}$

Range of Cosecant = $(-\infty, -1] \cup [1, \infty) = \mathbb{R} - (-1, 1)$

Range of Secant = $(-\infty, -1] \cup [1, \infty) = \mathbb{R} - (-1, 1)$

QUESTION

The number of solutions of the equation $\sin x = x^2 + 4x + 6$ is

- ☒ 1 One
- ☒ 2 Two
- ☒ 3 Three
- ☒ 4 Zero

(Sol) Range of LHS = $[-1, 1]$

$$\begin{aligned} \text{Here RHS} &= x^2 + 4x + 6 \\ \Rightarrow \text{RHS} &= (x^2 + 4x + 4) + 2 \\ \Rightarrow \text{RHS} &= (x+2)^2 + 2 \end{aligned}$$

$\Rightarrow$ Range of RHS = $[2, \infty)$

So the given equation has no solution

QUESTION

The number of solutions of the equation $2\cos x = x^2 + 2$ is

☒ One

☐ Two

☐ Three

☐ No solution

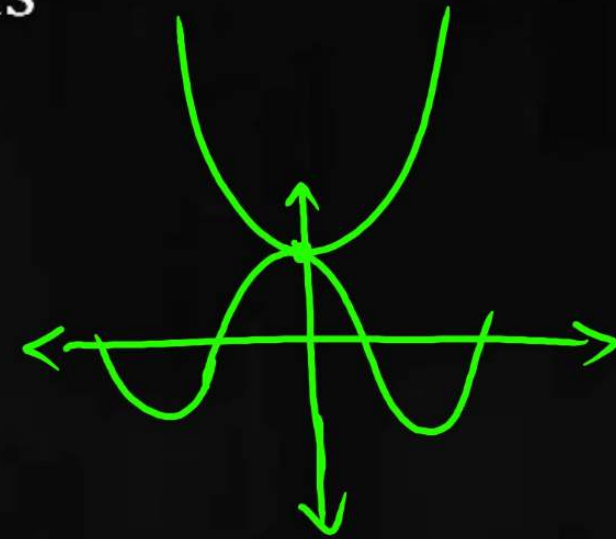
(Sol) Range of LHS = $[-2, 2]$

Range of RHS = $[2, \infty)$

$$\text{LHS} \Big|_{x=0} = 2 \times \cos 0 = 2 \times 1 = 2$$

$$\text{RHS} \Big|_{x=0} = 0^2 + 2 = 2$$

Desmos



QUESTION

The number of solutions of the equation $2\sin x = x^2 + 4x + 6$ is

- ☒ 1 One
- ☒ 2 Two
- ☒ 3 Three
- ☒ 4 Zero

(Sol) Range of LHS = $[-2, 2]$

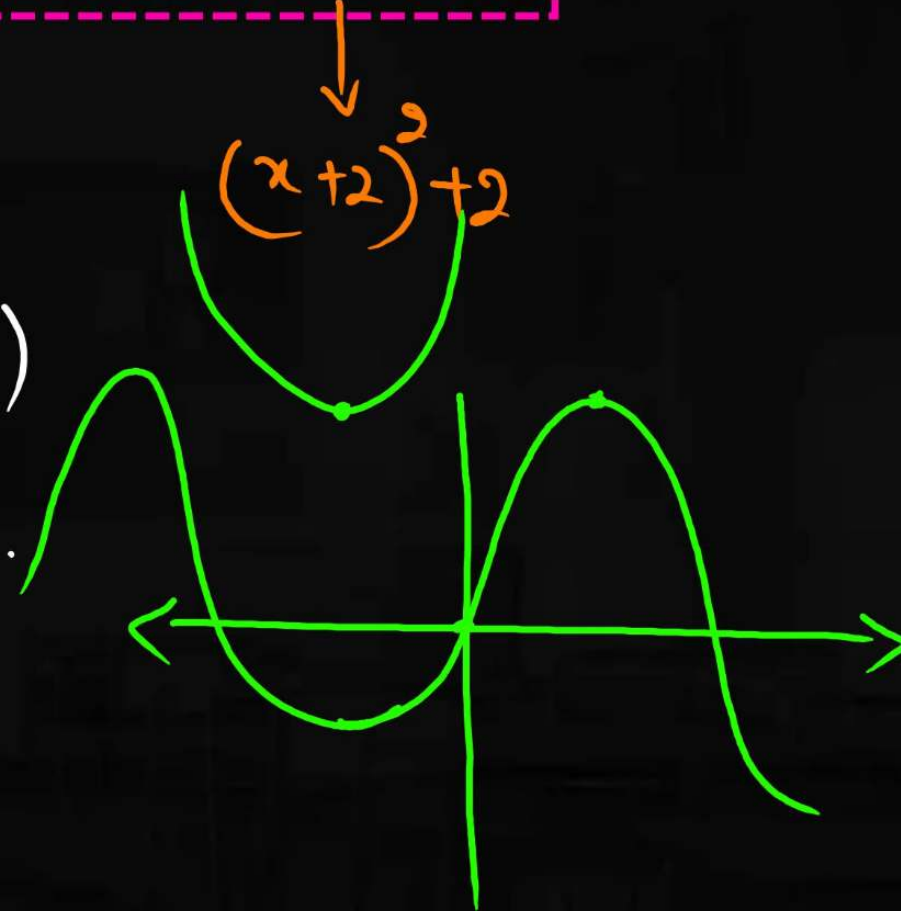
Range of RHS = $[2, \infty)$

Here LHS = $2 \times \sin \frac{\pi}{2} = 2$

$$x = \frac{\pi}{2}$$

RHS = $4 - 8 + 6 = 2$

$$x = -2$$



QUESTION (JEE-MAINS 2013)

The expression $\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A}$ can be written as

~~1~~ $\sin A \cos A + 1$

~~2~~ $\sec A \operatorname{cosec} A + 1$

~~3~~ $\tan A + \cot A$

~~4~~ $\sec A + \operatorname{cosec} A$

(Sol) $\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A}$
 $= \frac{\left(\frac{\sin A}{\cos A}\right)}{\left(1 - \frac{\cos A}{\sin A}\right)} + \frac{\left(\frac{\cos A}{\sin A}\right)}{\left(1 - \frac{\sin A}{\cos A}\right)}$
 $= \frac{\sin^2 A}{\cos A (\sin A - \cos A)} + \frac{\cos^2 A}{\sin A (\cos A - \sin A)}$

$= \frac{\sin^3 A}{\cos A \sin A (\sin A - \cos A)} - \frac{\cos^3 A}{\cos A \sin A (\sin A - \cos A)}$
 $= \frac{\sin^3 A - \cos^3 A}{\cos A \sin A (\sin A - \cos A)}$
 $= \frac{(\sin A - \cos A)^3 + 3 \sin A \cos A (\sin A - \cos A)}{\cos A \sin A (\sin A - \cos A)}$
 $= \frac{\cancel{(\sin A - \cos A)} ((\sin A - \cos A)^2 + 3 \sin A \cos A)}{\cos A \sin A \cancel{(\sin A - \cos A)}}$

$$= \frac{(\sin A - \cos A)^2 + 3\sin A \cos A}{\sin A \cos A}$$

$$= \frac{\sin^2 A + \cos^2 A - 2\sin A \cos A + 3\sin A \cos A}{\sin A \cos A}$$

$$= \frac{1 + \cancel{\sin A \cos A}}{\sin A \cos A \cancel{\sin A \cos A}}$$

$$= \operatorname{cosec} A \sec A + 1$$

$$= \sec A \operatorname{cosec} A + 1$$

$$x^2 + 4x + 6$$

$$= (\underline{x^2 + 4x + 4}) + 2$$

$$= (x + 2)^2 + 2$$

$$\downarrow$$

$$x^2 + 4x + 4$$

Sum Up:

Functions	Domain	Range
sine	$\mathbb{R}$	$[-1, 1]$
cosine	$\mathbb{R}$	$[-1, 1]$
tan	$\mathbb{R} - \{(2n + 1)\frac{\pi}{2} : n \in \mathbb{Z}\}$	$\mathbb{R}$
cot	$\mathbb{R} - \{n\pi : n \in \mathbb{Z}\}$	$\mathbb{R}$
sec	$\mathbb{R} - \{(2n + 1)\frac{\pi}{2} : n \in \mathbb{Z}\}$	$\mathbb{R} - (-1, 1)$
cosec	$\mathbb{R} - \{n\pi : n \in \mathbb{Z}\}$	$\mathbb{R} - (-1, 1)$

$$* \sin \theta = \sin(2n\pi + \theta), n \in \mathbb{Z}$$

$$* \cos \theta = \cos(2n\pi + \theta), n \in \mathbb{Z}$$

$$* \tan \theta = \tan(n\pi + \theta), n \in \mathbb{Z}$$

$$* \operatorname{cosec} \theta = \operatorname{cosec}(2n\pi + \theta), n \in \mathbb{Z}$$

$$* \sec \theta = \sec(2n\pi + \theta), n \in \mathbb{Z}$$

$$* \cot \theta = \cot(n\pi + \theta), n \in \mathbb{Z}$$

EXERCISE – 3.2

QUESTION



Find the values of other five trigonometric functions in Questions 1 to 5.

$$\cos x = -\frac{1}{2}, x \text{ lies in third quadrant.}$$

(Sol) Given that $\cos x = -\frac{1}{2}$

We know that $\cos^2 x + \sin^2 x = 1$

$$\Rightarrow \left(-\frac{1}{2}\right)^2 + \sin^2 x = 1$$

$$\Rightarrow \frac{1}{4} + \sin^2 x = 1$$

$$\Rightarrow \sin^2 x = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\Rightarrow \sin x = \pm \frac{\sqrt{3}}{2}$$

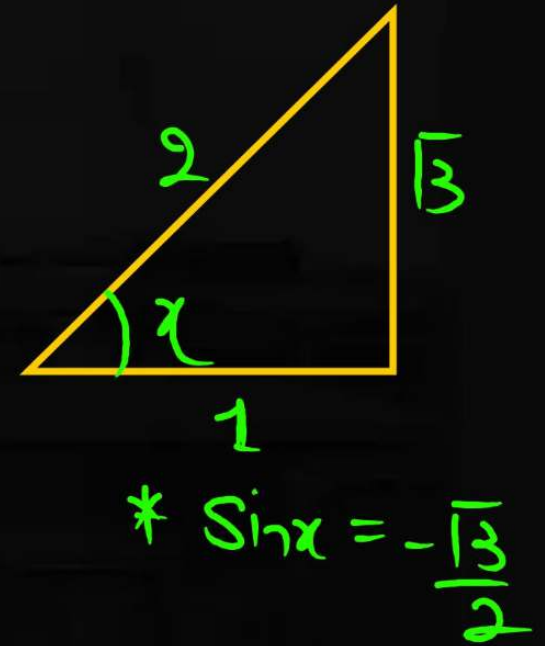
Since x lies in the 3rd Quadrant, so

$$\sin x = -\frac{\sqrt{3}}{2}$$

Also, $\tan x = \frac{\left(-\frac{\sqrt{3}}{2}\right)}{\left(-\frac{1}{2}\right)} = \sqrt{3}$

$$\cot x = \frac{1}{\sqrt{3}}, \sec x = -2$$

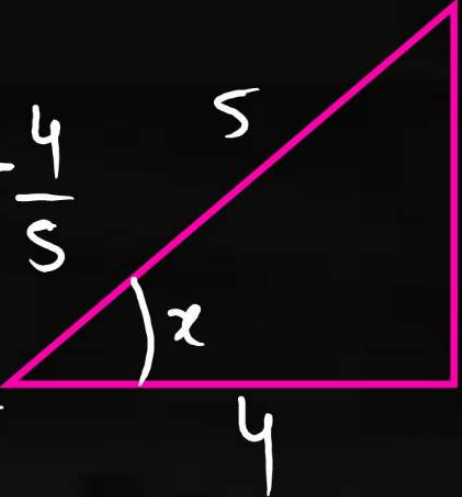
$$\operatorname{cosec} x = -\frac{2}{\sqrt{3}}$$



QUESTION

$$\sin x = \frac{3}{5}, x \text{ lies in second quadrant.}$$

(Sol) $\sin x = \frac{3}{5}, \cos x = -\frac{4}{5}$



$\csc x = \frac{5}{3}, \sec x = -\frac{5}{4}$

$\tan x = -\frac{3}{4}, \cot x = -\frac{4}{3}$

QUESTION



$\cot x = \frac{3}{4}$, x lies in third quadrant.



Homework
Problem

$$1 + \cot^2 x = \operatorname{cosec}^2 x$$

QUESTION



$\sec x = \frac{13}{5}$, x lies in fourth quadrant.

(Homework Problem)

QUESTION



$\tan x = -\frac{5}{12}$, x lies in second quadrant.

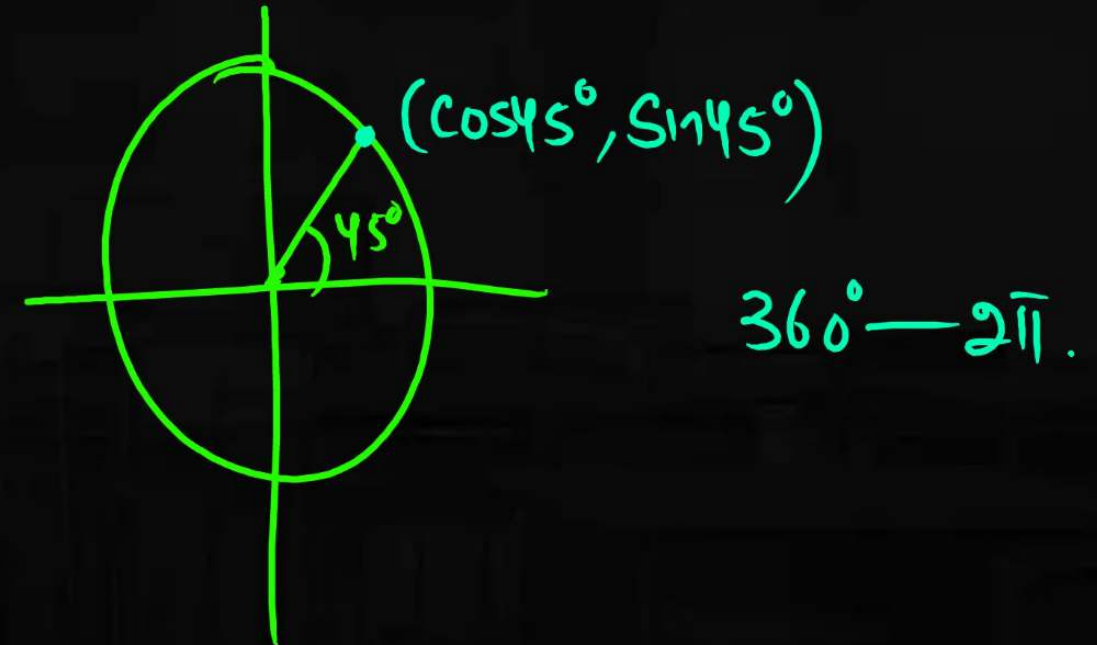
(Home work
Problem)

QUESTION

Find the values of ~~other five trigonometric functions in Questions 6 to 10.~~

$$\sin 765^\circ$$

$$\begin{aligned} \text{(Sol)} \quad \sin(765^\circ) &= \sin(2 \times 360^\circ + 45^\circ) \\ &= \sin 45^\circ = \frac{1}{\sqrt{2}} \end{aligned}$$



QUESTION

Find the values of ~~other five trigonometric functions in Questions 6 to 10.~~

$$\operatorname{cosec}(-1410^\circ)$$

$$\begin{aligned} \text{(Sol)} \quad \operatorname{Cosec}(-1410^\circ) &= \operatorname{Cosec}(4 \times -360^\circ + 30^\circ) \\ &= \operatorname{Cosec} 30^\circ \\ &= 2. \end{aligned}$$

$$\begin{aligned} \operatorname{Cosec}(2n\pi + \theta) &= \operatorname{Cosec} \theta \\ n &= -4 \end{aligned}$$

$$\operatorname{Cosec}(-8\pi + \theta) = \operatorname{Cosec} \theta$$

$$\Rightarrow \operatorname{Cosec}(4 \times -2\pi + \theta) = \operatorname{Cosec} \theta$$

QUESTION

Find the values of other five trigonometric functions in Questions 6 to 10.

$$\tan \frac{19\pi}{3}$$

$$(Sol) \tan\left(\frac{19\pi}{3}\right) = \tan\left(6\pi + \frac{\pi}{3}\right) = \tan\left(\frac{\pi}{3}\right) = \sqrt{3}.$$

$$\tan(n\pi + \theta) = \tan \theta.$$

QUESTION

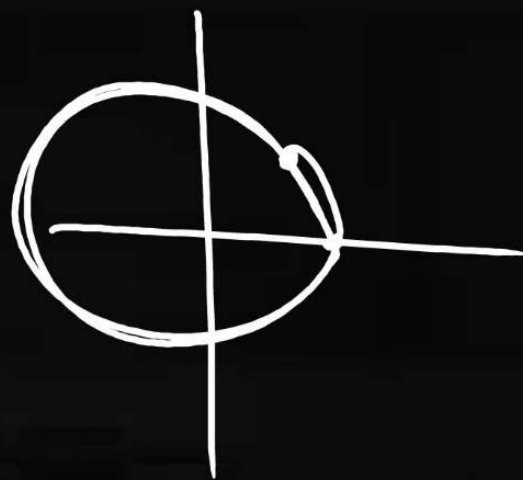


Find the values of ~~other five trigonometric functions in Questions 6 to 10.~~

$$\sin\left(-\frac{11\pi}{3}\right)$$

$$= \sin\left(-4\pi + \frac{\pi}{3}\right)$$

$$= \sin\left(\frac{\pi}{3}\right) = \frac{1}{2}$$



QUESTION



Find the values of ~~other five trigonometric functions in Questions 6 to 10.~~

$$\cot\left(-\frac{15\pi}{4}\right)$$

$$\cot(n\pi + \theta) = \cot \theta$$

$$= \cot\left(-4\pi + \frac{\pi}{4}\right) = \cot\left(\frac{\pi}{4}\right) = 1$$

Thank
You